June 19, 2026

Prof. Mathias Verbeke
KU Leuven
Leuven, Belgium

Dear Prof. Mathias Verbeke,

I am writing to express my strong interest in the PhD position on Causal Machine Learning for Industrial Root Cause Analysis (CausAICA project) at the M-Group, KU Leuven. Having reviewed the specific goals of uncovering the root causes of machine failures under limited data and evolving operating conditions, I am extremely motivated to bring my background in physics-informed AI and time-series modeling to your team.

My State Engineer thesis, "Gas Turbine Digital Twin," aligns perfectly with the industrial focus of this project. I engineered predictive maintenance architectures using ST-KDFormer and Temporal Convolutional Networks to predict thermodynamic degradation and sensor drift in highly complex, heavy machinery. While these deep learning models effectively captured sequential dependencies from multivariate time-series, I deeply recognized the critical limitation of pure, non-causal associative learning: it predicts *what* is failing, but struggles to mathematically isolate *why* it failed across heterogeneous operating environments.

To move beyond pure correlation, my Master's thesis ("Cardiovascular Medical Digital Twin") introduced PI-DeepONet, an architecture that explicitly constrained neural predictions to the causal 0D Windkessel differential equations via PyTorch Autograd. This allowed the model to merge raw sensor data with true, physics-based structural knowledge.

I am eager to transition this intuition into formal causal discovery and inference (e.g., Double Machine Learning, Causal Forests) for your project. Developing hybrid causal graphs that unite data-driven time-series extraction with structural physics-based models—precisely as outlined in the CausAICA framework—is the exact research path I wish to pursue. I am highly proficient in Python, have extensive experience with complex telemetry, and work with a high degree of independence.

Thank you for your consideration.

Sincerely,
Ismail Aissaoui
https://ismail-aissaoui.vercel.app

AISSAOUI Ismail

AI & Data Science Engineer

Email: ismail.aissaoui.pro@gmail.com   Phone: +213 660 70 77 96
Portfolio: ismail-aissaoui.vercel.app   Location: Chlef, Algeria
LinkedIn: linkedin.com/in/aissaoui-ismail-6a77a92a7   GitHub: github.com/i-aissaoui

PROFILE

AI & Data Science Engineer with a background in Full-Stack Software Engineering. Experienced in building projects across diverse domains, including Industrial Digital Twins, Healthcare, Financial Trading, Computer Vision, and Enterprise Management Systems. My primary interests include Deep Learning, Graph Neural Networks, Reinforcement Learning, and Natural Language Processing (NLP).

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL
ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, Lightning, ONNX
AI Techniques: Transformers (BERT, GPT, RAG), GNN, GAN, XGBoost, LSTM
Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization
Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS
Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy
Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra
Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

PROFESSIONAL EXPERIENCE

Graduate AI Research Engineer (Intern) — Jan 2026 – Jul 2026
Sonatrach (Industrial Surveillance Division) — Algeria
Architected an AI-native Digital Twin for real-time monitoring of Gas Turbines.

- Architected a 3-tier Dual-Track framework decoupling safety-critical Edge models from Cloud analytics.
- Developed **TCNEdge** achieving **0.049ms INT8 ONNX C++ inference** for emergency shutdown.
- Engineered **MambaTS** with a missingness-aware gate to maintain robustness during sensor dropouts.
- Implemented **KAN-PCNN** and **ST-KDFormer** to ensure strict thermodynamic consistency at scale.

Network Engineering Intern — Sep 2024
Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA) — 2021–2026
Engineering and Master's Degree in Computer Science
Specialization: Artificial Intelligence & Data Science

CERTIFICATIONS

DeepLearning.AI: Machine Learning Specialization, Deep Learning Specialization, NLP Specialization

ZTM Academy: PyTorch: Zero to Mastery, TensorFlow Developer

ACADEMIC THESES

State Engineering Degree Thesis:
Physics-Informed Explainable Digital Twin for Industrial Gas Turbines: A Dual-Track Cloud–Edge Deep Learning Framework with Koopman-Augmented State-Space Models

Master's Degree Thesis:
Hemodynamic Digital Twins: Fusing Hemodynamic Physics-Informed Neural Networks and Mamba-SSM for ICU Patient Prognostics

PROJECTS

Industrial Gas Turbine Digital Twin (Sonatrach) 2026

- **Edge & Cloud Architecture**: Engineered a 3-tier framework decoupling safety-critical TCNEdge (Siemens PLC) from near-edge MambaTS and cloud-based ST-KDFormer.
- **Ultra-Low Latency Inference**: Compiled 1D Dilated Convolutions (TCNEdge) to INT8 ONNX C++, achieving 0.049ms latency for emergency shutdown.
- **Physcis & Robustness**: Implemented MambaTS with missingness-aware gating for broken sensors, and KAN-PCNN with PyTorch Autograd for absolute thermodynamic consistency.

Tech: MambaTS, TCNEdge, ST-KDFormer, KAN-PCNN, PyTorch, C++ ONNX, PINN

Cardiovascular Medical Digital Twin 2026

- **High-Frequency Edge-Cloud Architecture**: Engineered **Medical_MambaTS** for sub-millisecond bedside inference and dual-branch **SST_KODA_MultiScale** for robust cloud forecasting using 100Hz VitalDB ICU telemetry.
- **Physics-Informed DeepONet**: Implemented a 0D Windkessel **PI_DeepONet** with PyTorch Autograd, forcing the model to obey cardiovascular fluid dynamics without retraining for new patients.
- **Data Pipeline & Robustness**: Developed a PyArrow Parquet zero-copy pipeline avoiding GPU starvation, and designed a custom missingness-aware gate to handle sensor dropouts natively.

Tech: PyTorch, Medical_MambaTS, PI_DeepONet, SST_KODA_MultiScale, Parquet PyArrow, Autograd

Aura — Enterprise AI Trading Platform 2025

- **Hybrid Forecasting**: Built an autonomous trading system combining LSTM networks for time-series predictions.
- **Strategic Decision-Making**: Implemented Deep Q-Learning (Reinforcement Learning) to optimize long-term reward policies.
- **Real-Time Dashboard**: Engineered a high-throughput FastAPI backend with WebSockets and a Next.js interactive UI.

Tech: TensorFlow, PyTorch, FastAPI, Next.js, WebSockets, RL

GNN-based Recruiting System (Career Connect) 2024–2025

- **Relational Learning**: Developed a recommendation engine using Graph Neural Networks (GNN) to map candidate-job relationships.
- **Semantic Embeddings**: Utilized PyTorch Geometric and SBERT for deep semantic text processing.
- **System Architecture**: Deployed via FastAPI and PostgreSQL, significantly outperforming keyword-based heuristics.

Tech: PyTorch Geometric, SBERT, FastAPI, PostgreSQL, GNN

LANGUAGES

Arabic: Native | English: Advanced | French: Professional

Engineering & Master's Thesis Summaries

Ismail Aissaoui
AI & Data Science Engineer

Master's Thesis: Cardiovascular Medical Digital Twin

Focus: Real-time Hemodynamic Inference, State-Space Models, Physics-Informed Neural Networks
Institution: Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)

This thesis proposes the design and implementation of a highly scalable, low-latency **Cardiovascular Medical Digital Twin** capable of performing sub-millisecond inference on high-frequency, multivariate medical telemetry. The core architectural contribution is the **Medical_MambaTS** framework, which leverages linear-time state-space models to solve long-sequence dependencies without the quadratic computational bottleneck of traditional Attention mechanisms.

To bridge the gap between purely data-driven black boxes and rigid mathematical models, the thesis introduces the **PI-DeepONet** (Physics-Informed Deep Operator Network). By integrating 0D Windkessel fluid dynamics equations directly into the PyTorch Autograd training loop, the architecture mathematically enforces physical consistency upon the cardiovascular predictions. Furthermore, addressing the clinical reality of disjointed hospital sensors, the system integrates the **SST-KODA-MultiScale** pipeline to robustly impute missing telemetry across non-uniform sampling rates, culminating in a resilient, physics-grounded clinical decision support tool.

State Engineer Thesis: Gas Turbine Digital Twin

Focus: Industrial Predictive Maintenance, Time-Series Forecasting, Thermodynamics
Institution: Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)

This thesis focuses on the development of an industrial-grade **Gas Turbine Digital Twin** designed for real-time sensor forecasting and thermodynamic state estimation. The primary objective is to prevent catastrophic failure in heavy machinery by anticipating structural anomalies and sensor drift across highly chaotic, non-stationary thermal environments.

The project benchmarks and deploys a spectrum of advanced sequence modeling architectures, including **LSTMs**, **Transformers**, **Temporal Convolutional Networks (TCN)**, and the hybrid **ST-KDFormer**. The implemented architectures were rigorously evaluated against strict thermodynamic constraints, ensuring that predictions regarding temperature gradients, exhaust velocities, and pressure metrics adhered strictly to the laws of energy conservation. By establishing a direct mapping between raw telemetry signals and impending mechanical degradation, this digital twin serves as a highly reliable predictive maintenance engine for large-scale energy infrastructure.

Academic References

Ismail Aissaoui

AI & Data Science Engineer

Reference 1

Dr. Mohammed Bedjaoui
Primary Thesis Supervisor
Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)
Email: m.bedjaoui@esi-sba.dz
Relationship: Supervised both my Master's Thesis (Cardiovascular Medical Digital Twin) and State Engineer Thesis (Gas Turbine Digital Twin). Can comprehensively vouch for my technical research capability, engineering architecture skills, and work ethic.

Reference 2

Dr. Belkacem Khaldi
Academic Professor
Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA)
Email: b.khaldi@esi-sba.dz
Relationship: Academic professor and course instructor. Can vouch for my theoretical understanding of complex systems and overall academic excellence.

Notice of Pending Graduation

Ismail Aissaoui

Please be advised that I am scheduled to officially defend my Master's and State Engineer theses and graduate on **June 29, 2026**.

My final, official diplomas and certified transcripts will be issued by the Higher School in Computer Science of Sidi Bel-Abbès (ESI-SBA) shortly thereafter.

The following pages contain my current academic transcripts to date.